Research proposal on attention selection mechanism

1. Title and Research question:

Title: Presynaptic residual Calcium Facilitation in Primary Visual Cortex as a Substrate for Rapid Attentional Selection

Research question: How do transient attentional signals navigate local networks structurally modified by Long-Term Potentiation (LTP) during long-term learning? Is presynaptic residual calcium facilitation at top-down axon terminals in V1 act the primary driver for the rapid recruitment of learned pyramidal-parvalbumin (PYR–PV) ensemble activation during attentional selection?

2. Background and Rationale

Calcium-mediated synaptic facilitation has previously been proposed as a substrate for short-term memory, in which residual calcium (Ca^{2+}) acts as a synaptic buffer that can be loaded, maintained, and read out by subsequent activity (Mongillo et al., 2008). Whether such a mechanism selects learned visual representations in primary visual cortex (V1) is unknown. The central hypothesis is that, after learning, residual presynaptic calcium transiently increases the efficacy of defined V1 synapses carrying task-relevant top-down or recurrent signals allowing attentional selection to preferentially recruit learned pyramidal excitatory neurons (PYR) and parvalbumin-expressing (PV) inhibitory interneurons subnetworks within a sub second window, without requiring the circuit to identify and assemble the appropriate pathway de novo on every trial.

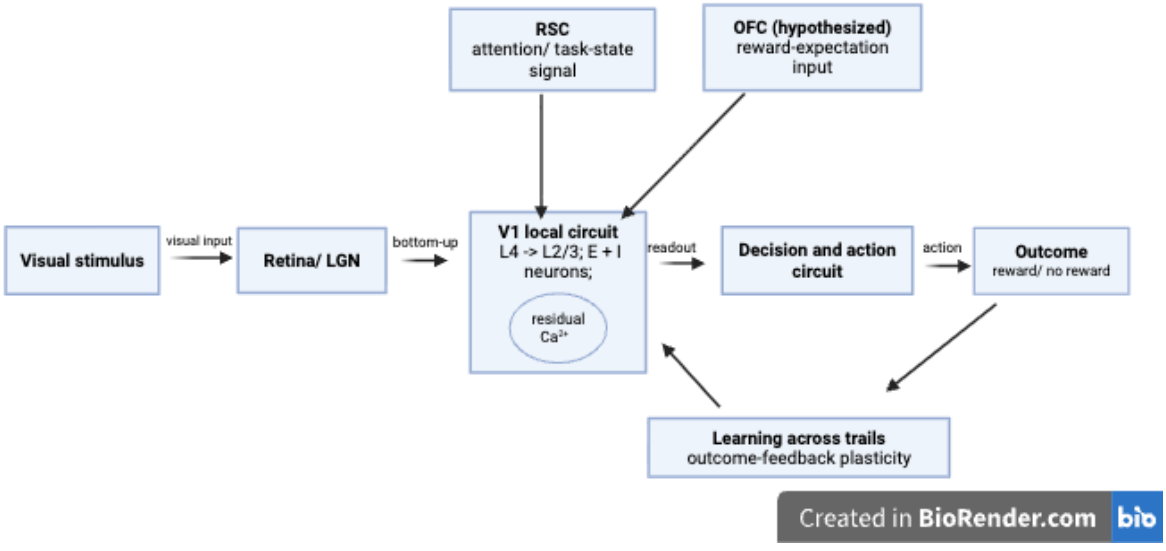


Figure 1: Lateral geniculate nucleus (LGN) processes the information received from retinal ganglion cells (RGC) and relays to the primary visual cortex (V1); V1 local circuit includes neurons from layer 4 (L4), layer 2/3 (L2/3) and excitatory-inhibitory (E-I) neurons, gated by top-down task state and sends refined representation to decision/action circuits modified across learning via outcome feedback mechanism. It stores the associations of the stimulus representations with action and outcome along with neural circuit representation for quick decision-making in the future.

3. Specific Aims & Hypotheses

Aim 1. Define the circuit signature of rapid attentional selection.

Hypothesis: Short-term attentional selection preferentially recruits pyramidal–parvalbumin (PYR–PV) ensembles formed during visual discrimination learning, where the magnitude and latency of ensemble recruitment directly predict visual discrimination performance.

Aim 2. Identify the locus of candidate calcium-dependent synaptic facilitation.

Hypothesis: Task-relevant short-term facilitation occurs primarily at top-down axon terminals in V1 rather than local recurrent PYR-to-PV connections, serving as the core mechanism for importing task intent rather than amplifying a learned local representation.

Aim 3. Determine the functional dependency of presynaptic residual calcium in rapid ensemble recruitment.

Hypothesis: Selective attenuation of presynaptic residual calcium will disrupt rapid, stimulus- and rule-specific ensemble recruitment while preserving the baseline visual responses or general task engagement. Furthermore, schizophrenia-relevant genetic model such as 22q11.2 deletion will exhibit altered presynaptic calcium kinetics and impaired calcium-dependent facilitation, directly causing abnormal recruitment of learned PYR–PV ensembles bridging the gap from this presynaptic dysfunction to impaired rapid visual attentional gating in V1.

4. Proposed Methodology

1. Learning task training and Ensemble imaging: Wild-type and mutant mouse cohorts will be trained on a visual discrimination task where distinct stimuli predict specific choices or outcomes (Poort et al., 2015, Poort et al., 2022). Repeated two-photon imaging of V1 pyramidal (PYR) and parvalbumin-positive (PV) interneurons across learning and long retention intervals will determine whether learned ensemble representations exhibit higher trial-to-trial reproducibility than novel stimuli along with accuracy in choice prediction (Poort et al., 2012, Khan et al., 2018).

2. Pre-synaptic Calcium Imaging: To measure cue-dependent transient Ca^{2+} loading in top-down projections, a Retrograde Adeno-Associated Virus expressing Cre recombinase (AAV-retro-Cre) vector will be injected into V1 to drive pathway-selective expression of a genetically encoded calcium indicator (GCaMP) in frontal or higher visual area neurons (Khan et al., 2018). For an attentional hypothesis, the axon boutons from the Retrosplenial Cortex (RSC) and Orbitofrontal Cortex (OFC) terminating in V1 will be imaged while recording local V1 activity and behaviour recording the exact timing and strength of top-down attentional signals.

3. Targeted Presynaptic Perturbation: To selectively perturb calcium-dependent facilitation in the projection of the relevant bouton population and time window identified can be genetically encoded to express slow-binding Ca^{2+}chelators inside the presynaptic axon boutons or knock out the Synaptotagmin-7 (Syt7) responsible for presynaptic short-term facilitation (Jackman et al., 2016, Jackman and Regehr, 2017). Presynaptic perturbation constructs will be validated in acute live brain slice preparations prior to in vivo longitudinal imaging and behavioural testing in mouse.

4. Expected Impact & Timeline Summary

This work will establish whether calcium-mediated synaptic facilitation in primary visual cortex ($V1$) serves as the cellular substrate for rapid attentional selection without requiring full network rewiring de novo on every trial. On top of this, the precise neural locus of task-relevant short-term facilitation distinguishing between top-down axon terminals in $V1$ versus local recurrent PYR-to-PV connections will be established. Importantly, by mapping the functional dependency of presynaptic residual calcium, this study will exhibit altered presynaptic calcium kinetics and impaired calcium-dependent facilitation causing abnormal recruitment of learned PYR–PV ensembles in schizophrenia-relevant genetic model (22q11.2 deletion). This will uncover the key bridge from presynaptic dysfunction to the abnormal recruitment of learned PYR–PV ensembles resulting in impaired rapid visual attentional gating in V1.

Timeline:

The experimental workflow will be executed in a 3-phase progression:

Phase 1: Behavioural Training and Ensemble Imaging

Phase 2: Pathway-Selective Calcium Imaging

Phase 3: Targeted Presynaptic Perturbation

References

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